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Practice article

A novel dual closed-loop control scheme based on repetitive control for grid-connected inverters with an LCL filter

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ARTICLE INFO

Article history:

Received 27 September 2016

Received in revised form

12 October 2017

Accepted 15 January 2018

Available online xxx

Keywords:

Grid-connected inverters

Harmonic suppression

Inductor-capacitor-inductor (LCL) filter

Repetitive control

Dual-loop control

ABSTRACT

Grid-connected inverters with LCL filters need high steady-state control accuracy, fast dynamic response performance, and strong robustness to guarantee the power quality. However, there are many problems in traditional control strategies that restrict improvements to control system performance, such as poor dynamic performance of traditional single-repetitive control, large ripples, low steady-state accuracy of inverter current feedback based repetitive dual-loop control or grid-current feedback based single-loop proportional-integral control. In this paper, a novel dual closed-loop repetitive control strategy based on grid current feedback is proposed for single-phase grid-connected inverters with LCL filters. The proportional-integral inner loop is stabilized by using an inherent one-beat delay achieved by digital controller. Based on the inner loop system, a detailed design scheme of a repetitive controller is presented, through which direct control of the grid current is realized, the reference is tracked perfectly to a zero phase shift, and high-attenuation gain is achieved in the high frequency range. In particular, the grid-voltage feed forward control and current reference feedforward control are adopted to suppress grid-voltage disturbance and increase dynamic tracking performance. Finally, the simulation and experimental results show that the proposed method has the advantages of high steady-state accuracy, fast dynamic response, and anti-disturbance ability.

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1. Introduction

Along with the development of new energy and renewable energy resources, distributed power generation systems (DPGS) have attracted a lot of interest in the past few years. As an interface equipment for DPGS, grid-connected inverters have been widely studied [1–4]. The grid-connected inverter, which is essentially a voltage-source inverter (VSI) with voltage input and current output, is the core of grid-connected power systems. The most important indexes for measuring the grid-connected inverter are total harmonic distortion (THD) of the grid current and the grid power factor (PF) [5,6].

In order to reduce the switching ripples and obtain a low grid-current THD, L filters and LCL filters are mainly used as grid-connected inverters. Compared with L filters, LCL filters have better filtering performance because the additional capacitor branch can bypass high-frequency harmonics [7]. However, the three-order system composed of an LCL filter increases the difficulty of the controller design, and the resonance peak caused by the LCL filters severely affects the system stability [8,9]. Systems with LCL filter present lower harmonic impedance, and therefore they are susceptible to background harmonic interference. As a result, for grid-connected inverters based on the LCL filter, reliable current tracking is not only required but also a strong ability to suppress the grid impedance variations and background harmonics are expected.

To address this issue, numerous current control schemes have been proposed for LCL-type grid-connected inverters, such as proportional-integral (PI) control, proportional-resonant (PR) control, deadbeat control, repetitive control (RC), etc. Traditional PI control, which has a simple structure, is easy to implement, but it

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has limitations for harmonic suppression, and the stability margin of the constructed system is not high [10,11]. PR control has the capability of obtaining a high gain and can perfectly track the sinusoidal reference signal [12]; however, every regulator for each harmonic frequency has to be adopted in the method, which increases the computation quantity and the complexity of the system. Moreover, in order to maintain good performance, all of the resonant frequencies have to be identical to the grid fundamental and harmonic frequencies [13]. Deadbeat control has a higher bandwidth and faster dynamic responses, but it relies heavily on mathematical models. Therefore, disturbances on *LCL* filter should be fully considered in the design [14].

RC [15] has been widely used in the control of grid-connected inverters [16–24], stand-alone inverters [25], and active filters [26,27] due to its excellent tracking performance. Using state feedback, which was proposed in Ref. [16], to configure the controlled object can improve system stability, and enhance the system dynamic-response ability, but more feedback will increase the cost of the system. A single-loop RC proposed in Refs. [17] and [18] has good steady-state performance that can be achieved by a frequency-adaptive mechanism. However, it cannot meet renewable energy requirements on dynamic response. In Ref. [19], the compensation performance is increased by introducing the IIR filter to the RC, but the dynamic performance of the system cannot be improved. To enhance the dynamic response, a variety of hybrid controls are widely used. RC and multi-resonant control (MRC) schemes are used in multifunctional grid inverters [20] and [21], which can provide a large gain at selective frequencies and offer superior tracking ability for both the fundamental and harmonic signals. However, the system is more complex, and there is a coupling problem between the MRC and the RC. A hybrid control scheme combined with PI and RC is presented in Refs. [22] and [23], in which the PI controller and the RC in parallel can carry out their duties. The PI controller ensures fast dynamic response of the grid-connected inverters, and the RC is used as a harmonic compensation to increase system stability. Dual-loop control based on the inverter-current feedback applied in Refs. [22] and [23] has fast dynamic response, but grid-connected currents cannot be controlled directly, and the grid-connected power factor cannot be guaranteed. As a result, the steady-state accuracy is poor, and the potential oscillation tendency of the grid-side inductor and filter capacitor cannot be suppressed. Although these methods can improve the dynamic performance, they are not ideal for grid harmonic suppression and anti-impedance disturbance.

A novel repetitive dual-loop control scheme of a grid-connected inverter with an *LCL* filter is proposed in this paper to realize precise control of grid-connected inverters. This inverter is composed of a PI inner loop and RC outer loop based on grid-connected current feedback. The inner loop unit is corrected to a zero phase shift system by the PI controller, and the stability control of the *LCL* filter is realized by using the inherent one-beat delay characteristic of digital control. Consequently, the inner-loop structure is extremely simple because it uses only one feedback quantity. Based on the inner loop unit, the inner model and compensator of the RC are designed. In particular, the effects of voltage fluctuations and distortion on the stability of the power grid are fully considered, so the grid-voltage feedforward control is introduced to suppress the voltage harmonics. Meanwhile, for the dynamic delay of the RC outer loop, current reference feedforward is increased. When the reference current is suddenly changed, the current fast track is realized by the PI inner loop before the RC has not been put into action, and the dynamic response capability is enhanced, which makes up the deficiency of the RC outer loop. The designed controller will realize the direct control of the grid current, track perfect reference with zero phase shift, and achieve extremely

high-attenuation gain at a high-frequency range. The experiment shows that the proposed control system has a high steady accuracy and an extremely strong harmonic-suppression capability. The rest of this paper is organized as follows:

The overall structure of the system is described in Section 2. In Section 3, the proposed control method in this paper is discussed in detail. The controller performance is analyzed in Section 4. In Section 5 the simulation and experimental results of the control system are presented, which validates the proposed method. Conclusions are stated in Section 6.

2. Description of the system

The proposed control system is shown in Fig. 1. The main circuit topology is a single-phase grid-connected inverter with *LCL* filter. The repetitive dual-loop control method is adopted. The outer loop is controlled by the RC, which makes the grid-connected current i_g track the sinusoidal reference i_{ref} without a steady-state error. The PI control method is applied in the inner loop, which can increase the damping of the system to suppress the resonance peak. The output of the repetitive controller and the reference current are taken simultaneously as the reference of the inner loop to increase dynamic response ability of the system and to eliminate the effect of the RC one-cycle delay. The pulse width modulation (PWM) signal is derived from the output of the dual-loop structure. The grid-voltage feedforward is introduced into the control system, which can eliminate the voltage disturbance and decrease the burden on the controller. To synchronize the grid-connected current i_g with the grid voltage v_g , the grid-current reference i_{ref} is generated using the grid-voltage phase angle θ detected by a digital phase-locked loop (DPLL) and the given current amplitude I_g^* . A DPLL is used to not only provide the phase angle θ of the current reference but also calculate the ratio N of the sampling frequency f_s to the grid frequency f_0 for the RC. It is assumed that the inverter is powered by a constant dc-voltage source. Under this condition, the controller does not need to adjust the dc-link voltage. In this method, the grid voltage and grid current only need to be measured in real time.

3. Design of the repetitive dual-loop controller based on grid-connected current feedback

As reported in the literature, RC has a good steady performance, but the internal mode regulated by the period of the grid voltage will result in a long dynamic response process. The dynamic response of the PI control based on the grid-connected current is

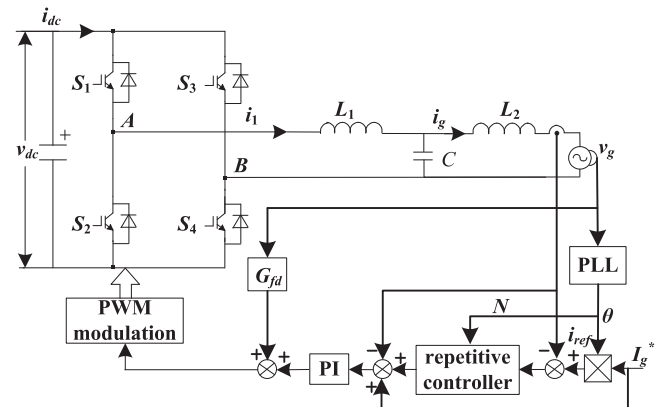


Fig. 1. Block diagram of a single-phase grid-connected inverter with the repetitive dual-loop control scheme.

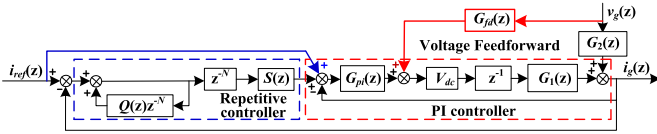


Fig. 2. Block diagram of repetitive dual-loop controlled grid-connected inverters with grid current feedback.

very fast, but the control accuracy is not high. As a result, the combination of the two methods can improve the stability and dynamic performance of the system at the same time.

The repetitive dual-loop control method is proposed in this paper. The method's topology is composed of a PI inner loop and RC outer loop, as shown in Fig. 2. Due to the existence of the periodic delay, the repetitive controller cannot respond to the changed reference in the first fundamental period. However, the inner loop can quickly respond to the feed-forward current, which guarantees the dynamic performance of the system. In the second fundamental period, the RC is only required to correct the tracking error on the basis of the inner loop. Therefore, the adjustment process can be significantly shortened, and the fast-tracking performance of the system is also enhanced. Furthermore, the grid-voltage feedforward is introduced into the system to increase the ability to suppress grid-voltage disturbances.

In Fig. 2, $G_1(z)$ is the transfer function of the LCL Filter. $G_2(z)$ is the transfer function of the grid voltage to the grid current. $G_{pi}(z)$ is the transfer function of the PI controller of the inner loop. $Q(z)$, z^{-N} , and $S(z)$ are three factors of the RC scheme, respectively. $G_{fd}(z)$ is the transfer function of the grid-voltage feedforward. $v_g(z)$ is grid voltage. $i_g(z)$ is the output current of the grid-connected inverter injected into the grid, and $i_{ref}(z)$ is its reference. v_{dc} is the dc-link voltage. z^{-1} is the one-beat delay factor inherent in digital controls. The specific design of the control method is described as follows.

3.1. Model of the single-phase grid-connected inverter based on an LCL filter

The power unit consists of an inverter bridge, an LCL filter composed of an LC filter (L_1 and C) and a grid interface inductor L_2 , where the parasitic resistances of the inductor are ignored, as was shown in Fig. 1. The structure diagram of an LCL filter is shown in Fig. 3. $G_1(s)$ and $G_2(s)$ can be expressed as follows.

$$G_1(s) = \frac{i_g(s)}{u_{in}(s)} = \frac{R_c Cs + 1}{L_1 L_2 Cs^3 + (L_1 + L_2) R_c Cs^2 + (L_1 + L_2)s} \quad (1)$$

$$G_2(s) = \frac{i_g(s)}{v_g(s)} = \frac{1}{L_1 L_2 Cs^3 + (L_1 + L_2) R_c Cs^2 + (L_1 + L_2)s} \quad (2)$$

$G_1(s)$ and $G_2(s)$ are the transfer functions of the input voltage u_{in} of the LCL filter and the grid voltage v_g responds to the grid current i_g respectively. The parameters of the LCL filter are mainly designed according to [27], and the system parameters are given in Table 1.

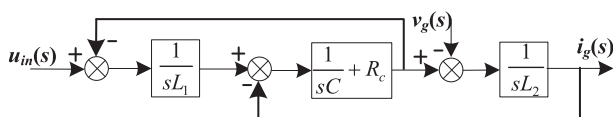


Fig. 3. Block diagram of LCL filter.

Table 1
Parameters of the grid-connected inverter.

Symbol	Description	Value
L_1	filter inductor	2 mH
C	filter capacitor	10 μ F
L_2	grid interface inductor	0.7 mH
R_c	parasitic resistance of filter capacitor	0.005 Ω
v_g	grid voltage	220 V
v_{dc}	DC bus voltage	400 V
f_0	ideal grid voltage frequency	50 Hz
f_s	sampling frequency	9 kHz
p	Output power	3.3 kW

3.2. Design scheme of the inner loop unit

The block diagram of the inner loop unit is shown in Fig. 4(a). In this paper, the carrier wave gain is selected as 1, so v_{dc} is the gain of the average switching model of the inverter. $G_{pi}(s)$ is the transfer function of the PI controller in the continuous system ($K_p = 0.025$, $K_i = 10$). The open-loop transfer function $G_{op}(s)$ shown in Fig. 4(a) can be written as

$$G_{op}(s) = v_{dc} G_{pi}(s) G_1(s) = G_{pi}(s) G_p(s) \quad (3)$$

The Bode diagram of the inner loop unit in the continuous system is shown in Fig. 5. It has been shown that the 83.7 dB resonance peak is caused by the series resonance at the 2.21 kHz. From b-b' in Fig. 5, after correction of the PI controller, the system is unstable because the 51.7 dB resonance peak still exists. To address this issue, the delay correction or low-pass filter can be used to attenuate the resonant peak from 51.7 dB to 0 dB; however, the low-frequency range gain will be greatly reduced, and the system steady-state accuracy will be severely weakened.

From b-b' in Fig. 5, the system gain is below 0 dB in the range from 650 Hz to 1.82 kHz. It can be seen that if the open-loop phase shift is increased to make the phase crossing point fall in the band, then a positive amplitude gain should be obtained so that the system will be stable. In fact, digital control has a control delay and zero order hold (ZOH) characteristic. Due to the delays in sampling and calculation, the loaded value of the modulated signal lags behind the calculated value of one sampling period. In addition, the modulated signal values are loaded at the start of each sampling period and remained unchanged throughout the sampling period. To describe this feature accurately, a delay link e^{-sT_s} and the ZOH are added to the control model. The ZOH function can be expressed as

$$G_{zoh}(s) = \frac{1 - e^{-sT_s}}{s} \quad (4)$$

In order to be closer to the actual digital control system, the continuous PI controller should be discretized by the ZOH method. The ZOH and the plant are connected in a series, and then discretized by the step response method. The block diagram of the inner loop unit in discrete system is shown in Fig. 4(b). The plant, open-loop and closed-loop transfer functions of the inner loop unit will be obtained, respectively, as

$$G_p(z) = v_{dc} G_1(z) \quad (5)$$

$$G_{op}(z) = z^{-1} G_p(z) G_{pi}(z) \quad (6)$$

$$G_{cl}(z) = \frac{G_{op}(z)}{1 + G_{op}(z)} \quad (7)$$

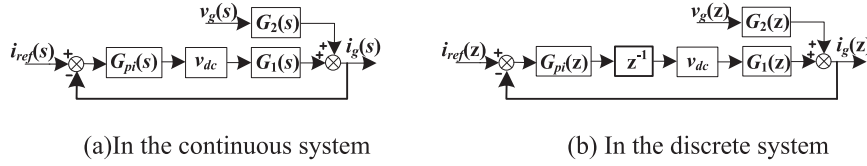


Fig. 4. Block diagram of the inner loop.

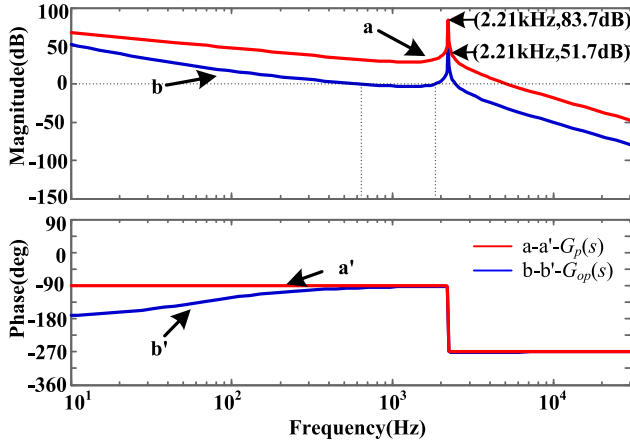


Fig. 5. Bode diagram of the inner loop in the continuous system.

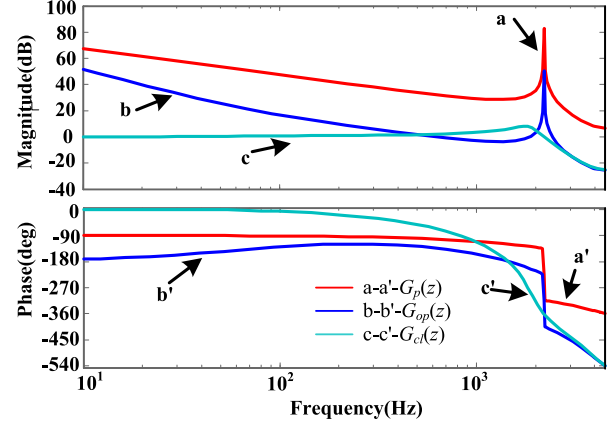


Fig. 7. Bode diagram of the inner loop in the discrete system.

The PI controller determines the stability and steady state accuracy of the inner loop. According to the model optimization method [28], K_p and K_i can be estimated using the principle of zero-pole cancellation and bandwidth optimization. Finally, using the Bode diagram analysis, PI controller parameters can be obtained according to the system stability margin and the control bandwidth. Fig. 6 shows the Bode diagram of the inner loop when K_p or K_i changes. As shown in Fig. 6, the delay characteristic of the digital system can result in a positive amplitude margin in the inner loop.

Fig. 6 (a) shows that when $K_p = 0.01$, K_i does not substantially affect the amplitude margin, but the larger the K_i is, the smaller the phase margin is, which is disadvantageous to stability. On the other hand, the control bandwidth and amplitude gain are relatively increased when K_i increases, which is advantageous for improving

the steady-state accuracy. Fig. 6 (b) shows that when $K_i = 10$ and $K_p = 0.05$, the amplitude margin is less than zero, and the inner-loop stability condition is not satisfied. When K_p changes from 0.01 to 0.03, K_p has a greater effect on amplitude margin. It can be seen that the larger K_p is, the larger the control bandwidth and amplitude gain is and the smaller the amplitude margin is. As a result, the choice of K_p and K_i is contradictory between stability and steady-state accuracy. In order to achieve a greater stability margin and control bandwidth at the same time, this paper carries on the compromise, using $K_p = 0.025$ and $K_i = 10$. The system inner loop correction results are shown in Fig. 7.

Fig. 7 shows the Bode diagram of the inner loop in the discrete system when the sampling frequency is 9 kHz. From b-b' in Fig. 7, it can be seen that the phase crossing frequency is 1.46 kHz, the amplitude margin is 3.49 dB, and the phase margin is 45°.

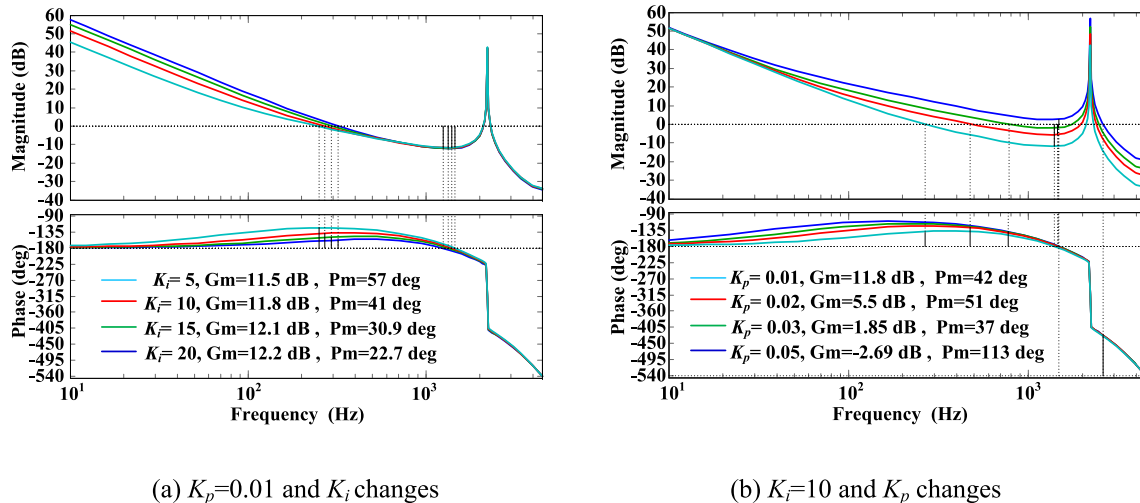


Fig. 6. Bode diagram of the inner loop when K_p or K_i changes.

Consequently, the control system is stable. The resonant peak still exists, but it is smaller than that of the original system. From c-c' in Fig. 7, the system can completely track the reference and achieve zero phase shift in the low frequency range. However, the phase in the high frequency range is slightly lagging. When the frequency is higher than 200 Hz, the input and output amplitudes are almost the same, but the phase lag increases gradually, which will seriously affect the control effect of the system. In order to achieve the goal of zero phase shift and zero gain, and to suppress resonant peak, a reasonable control outer loop unit should be added to the system.

3.3. Design of the outer loop

RC is applied in the outer loop of the proposed system. RC is a control strategy based on the internal-model principle [15,24], which can realize a zero steady-state error to track the input signal by the periodic accumulation on the input signal. At the fundamental and harmonic frequency of the control object, RC has a high gain that can compensate the phase and amplitude of the system, but also can suppress the disturbance of power grid impedance. Based on the sinusoidal characteristics of a given signal, the internal model of the repetitive controller $G_{im}(z)$ is chosen as

$$G_{im}(z) = \frac{1}{1 - Q(z)z^{-N}} \quad (8)$$

where $N(N=f_s/f_0)$ is sampling time in one period and $Q(z)$ can be chosen as a low-pass filter or a constant, which is slightly less than 1. $Q(z) = 0.95$ is selected in this paper.

As shown in Fig. 8, the inner model of the RC has a high gain and zero phase shift at the fundamental frequency and the multiples of the fundamental frequency. It follows that the internal model of RC has very good stability. However, in the inner loop unit, phase shift and low attenuation rate in the high frequency range after PI correction is a problem. Therefore, a compensation function should be added into the RC. A typical RC system block diagram is shown in Fig. 9. $G_c(z)$ is the plant, which is a closed-loop transfer function of the inner loop in this paper, and $Q(z)z^{-N}$ is the transfer function of the positive feedback loop of the inner model. It is essential to use the periodic delay link z^{-N} to correct the system phase. The

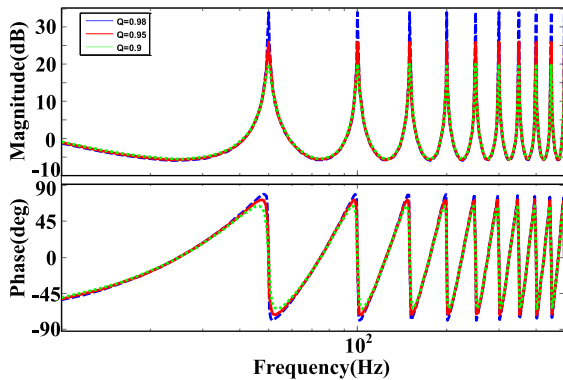


Fig. 8. Bode diagram of the inner model of the RC.

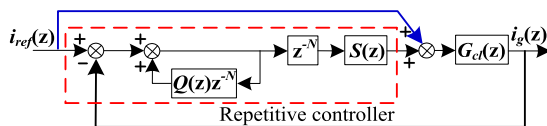


Fig. 9. Block diagram of a typical RC system.

amplitude and phase angle of the plant can be corrected by compensator $S(z)$, which determines the performance of the repetitive controller.

In order to achieve zero-phase compensation, the compensator $S(z)$ is preferably designed as the inverse of $G_c(z)$. However, $G_c(z)$ contains unmodeled dynamics and parametric uncertainties in practice, so it is not possible to achieve a completely accurate inverse model. Also, the inverse model may have poles outside the unit circle, which will result in system instability. Therefore, $S(z)$ is designed according to the structure features and frequency characteristics of $G_c(z)$. As a result, the corresponding notch filter, the second-order filter and the lead-link are designed as the compensator $S(z)$. From c-c' in Fig. 7, because the system achieves zero phase shift in the low frequency range, it is not necessary to compensate for phase. In the medium-frequency range, the phase-shift is gradually increased, but it can be compensated by the lead-link. The resonance peak is suppressed by the corresponding notch filter. In the high-frequency range, the amplitude gain of the inner loop unit can be attenuated by the second-order low-pass filter, while the total phase lag can be compensated by lead correction. With this method, the system has good steady-state performance.

3.3.1. Suppression of resonance peak

The function of the notch filter [29] is

$$F_1(z) = \frac{z^m + 2 + z^{-m}}{4} \quad (9)$$

where m is the order of the notch filter.

(9) can be written as:

$$F_1(\theta) = \frac{e^{jm\theta} + 2 + e^{-jm\theta}}{4} = \frac{\cos m\theta + 1}{2} \quad (10)$$

According to the design criterion of the notch filter, the first notch point is the resonance point in the medium-frequency range. Let $F_1(\theta) = 0$, and the following equation is:

$$m = \frac{\pi}{T_s \omega_0} \quad (11)$$

where $T_s = 1/f_s$ is the sampling period and ω_0 is the resonant frequency ($\omega_0/2\pi = 2.21$ kHz). When the sampling frequency is 9 kHz, m is 2.036. After taking the integer, m is 2, so the notch filter of the control system is designed as

$$F_1(z) = \frac{z^2 + 2 + z^{-2}}{4} \quad (12)$$

3.3.2. Attenuation of high frequency harmonics

The notch filter can only produce significant attenuation in the notch point. In order to achieve the obvious attenuation in the high frequency range, a second-order low-pass filter is needed to enhance the ability of the high frequency suppression. According to the frequency characteristics of the inner loop unit and the engineering requirements, the corner frequency ω of the filter is selected as $2\pi \times 2.5 \times 10^3$ rad/s and the damping ratio ξ is selected as 0.707. With these considerations, the filter is finally designed as

$$F_2(z) = \frac{0.6291z + 0.2634}{z^2 - 0.1922z + 0.08476} \quad (13)$$

3.3.3. Phase compensation

A fourth-order lead link is applied to compensate for the phase lag of the inner loop unit and the second-order low-pass filter. As can be seen from Fig. 10, after the fourth-order lead link compensation, the system has a zero phase shift characteristic. Therefore, this paper selects the fourth-order lead link for phase compensation.

As described above, the compensator $S(z)$ is considered as

$$S(z) = z^4 F_1(z) F_2(z) = \frac{0.1573z^5 + 0.06585z^4 + 0.31455z^3 + 0.1317z^2 + 0.1573z + 0.06585}{1 - 0.1922z^{-1} + 0.08475z^{-2}} \quad (14)$$

Fig. 11 shows the frequency characteristics of the inner loop unit after compensation. Fig. 11 shows that the resonant peak has been greatly reduced after the notch filter is used. However, after the compensation of the second-order low-pass filter, the low frequency and medium frequency characteristics are basically unchanged, and the attenuation is obviously increased at the high frequency range. As a result, the fourth-order lead link can perfectly compensate for phase lag. In short, through this compensation, the system has a zero phase shift characteristic at medium and low frequencies, and a high gain attenuation characteristic in the high frequency range.

To summarize, the closed-loop transfer function of the repetitive dual-loop controller has been obtained as

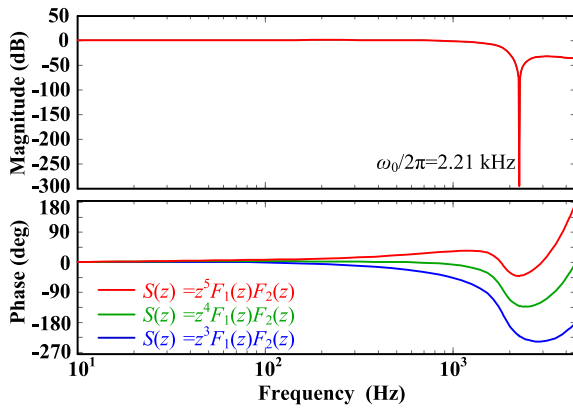


Fig. 10. Frequency characteristics of $S(z)G_{cl}(z)$ in different lead-link.

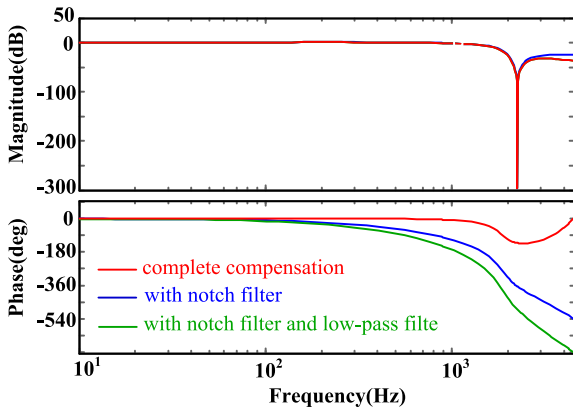


Fig. 11. Frequency characteristic of the system after compensation.

$$G(z) = \frac{G_{cl}(z)(RC(z) + 1)}{1 + RC(z)G_{cl}(z)} \quad (15)$$

where $RC(z) = z^{-N}S(z)/(1-Q(z)z^{-N})$. Fig. 12 shows the frequency characteristics of the repetitive dual-loop controller. From Fig. 12, in the fundamental frequency and medium and low frequency ranges, the system has a high open-loop gain up to 26 dB, closed-loop

characteristics close to zero gain, and zero phase shift. In the high frequency range, the system gain is very small, so it is obvious that the system has strong attenuation ability. Therefore, through inner-loop and outer-loop correction, a zero steady-state error can be achieved to track the reference and suppress the grid disturbance.

3.4. Application of the grid-voltage feedforward

Fig. 4 shows the block diagram of the inner loop unit without the grid-voltage feedforward. From Fig. 4, if a large number of harmonics are introduced into the grid voltage v_g due to the connected nonlinear load, it will seriously affect the quality of the grid current and the stability of the grid-connected inverter system. To solve this problem, an additional path is introduced into the output of $G_1(z)$, as shown in Fig. 13(a), which is the opposite effect of the original link to eliminate the grid-voltage distortions. But the new link is still not included in the forward channel, which means that the controller cannot suppress the grid-voltage disturbance. By moving the feedforward node from the output of $G_1(z)$ to the output of $G_{pi}(z)$, and modifying the feedforward function appropriately, the equivalent block diagram can be obtained as shown in Fig. 13(b). From Fig. 13(b), the grid-voltage feedforward can also suppress current harmonics caused by grid-voltage distortions. The function $G_{fd}(z)$ of the grid-voltage feedforward loop can be written as

$$G_{fd}(z) = \frac{zG_2(z)}{v_{dc}G_1(z)} = \frac{G_2(z)}{G_p(z)} \quad (16)$$

$$G_{fd}(z) = z \frac{0.002497z^2 + 0.00879z + 0.002495}{z^2 + 3.5154z + 0.9969} \quad (17)$$

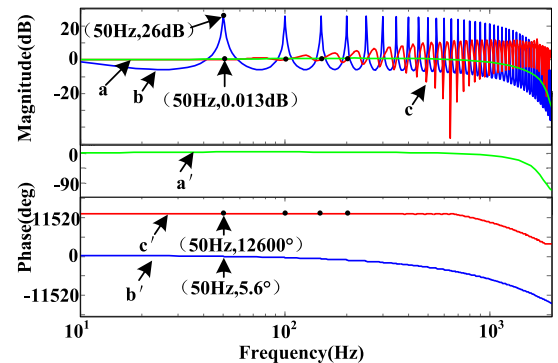


Fig. 12. Frequency characteristic of repetitive dual-loop controller.

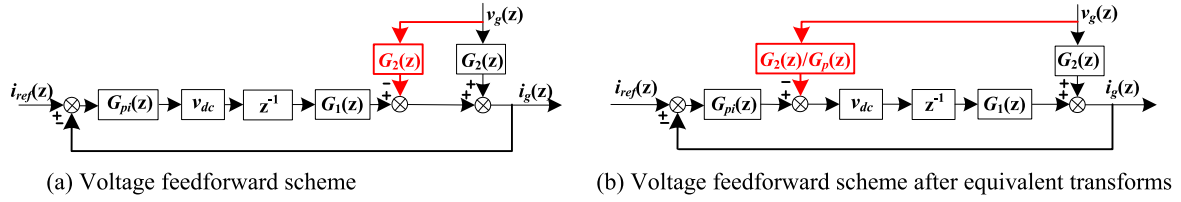


Fig. 13. Block diagram of feedforward scheme and its equivalent transforms.

According to the system parameters and transfer function mentioned above, the specific feedforward function (17) can be obtained. From (17), the present output of the feedforward control loop is not only related to the present grid voltage value, but also to the grid voltage value from the last two-beats. z is achieved by replacing the value of the current moment with the value of the next moment in the last period. This also reflects that the grid voltage was involved in the feedback control.

4. Performance analysis of the controller

Based on the analysis mentioned above, the relationship between output error, reference, and disturbance can be obtained from the system structure diagram as follows

$$e(z) = \frac{(1 - G_{cl}(z))(1 - Q(z)z^{-N})}{1 - z^{-N}(Q(z) - S(z)G_{cl}(z))} i_{ref}(z) + \frac{(1 - Q(z)z^{-N})(G_2(z) - G_{fd}(z)G_p(z))}{1 - z^{-N}(Q(z) - S(z)G_{cl}(z))} v_g(z) \quad (18)$$

4.1. Analysis of system stability

From (18) and according to RC characteristics, the sufficient condition for the stability of the system is derived as [26]

- 1) Inner loop transfer function $G_{cl}(z)$ is stable; namely, the roots of $1 + G_{op}(z) = 0$ are inside the unit circle.
- 2) The roots of the characteristic equation $1 - z^{-N}(Q(z) - S(z)G_{cl}(z)) = 0$ are inside the unit circle, and

$$|Q(e^{j\omega T}) - S(e^{j\omega T})G_{cl}(e^{j\omega T})| < 1, \omega \in [0, \pi/T_s] \quad (19)$$

Fig. 14 shows the pole distribution of $G_{cl}(z)$. Fig. 14 shows that the poles, that is, the roots of $1 + G_{op}(z) = 0$, are inside the unit circle. This indicates that the inner loop is stable. Fig. 15 shows the trajectory of $S(e^{j\omega T})G_{cl}(e^{j\omega T})$ for different inductors. When $Q(e^{j\omega T})$ is 0.95, it is obvious that the trajectory is always within the unit circle, with (0.95, 0) as the axle center. So, (19) is also satisfied. Moreover, the larger the amplitude of $S(e^{j\omega T})G_{cl}(e^{j\omega T})$ is, and the closer $Q(e^{j\omega T})$ is to 1, the smaller the steady-state error $e(z)$ is. From Fig. 11, it can be seen that the gain of $S(e^{j\omega T})G_{cl}(e^{j\omega T})$ remains basically the largest value at 0 dB, so the steady-state error $e(z)$ is very small. However, when $Q(e^{j\omega T})$ is closer to 1, the system stability is worse, as shown in Fig. 16. When $Q(e^{j\omega T}) = 1$, the system poles are beyond the unit circle, and the system is unstable. So $Q(e^{j\omega T})$ was selected as 0.95 in compromise.

The influence of the PI parameter on system stability, which is the closed-loop pole diagram when K_p and K_i are changed, is shown in Fig. 17. As shown in Fig. 17(a), when K_p changes from 0.01 to 0.1, the repetitive control poles are close to the boundary of the unit circle, and when the K_p is bigger than 0.04, the system poles have moved outside the unit circle. As shown in Fig. 17(b), when K_i

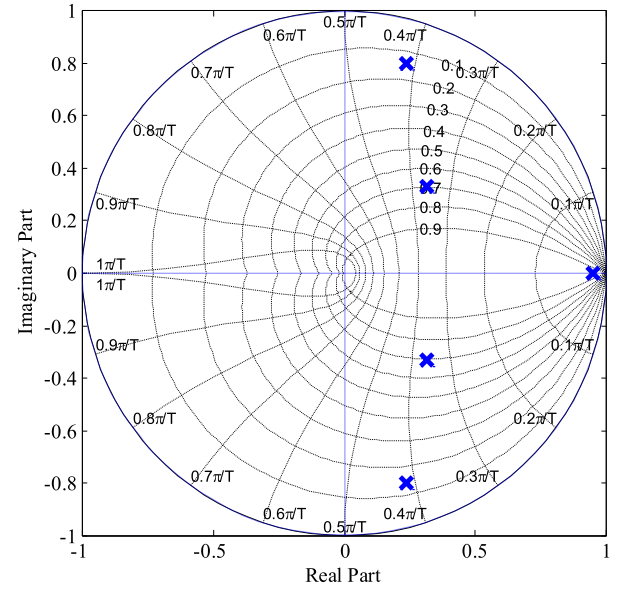


Fig. 14. Pole diagram of inner loop.

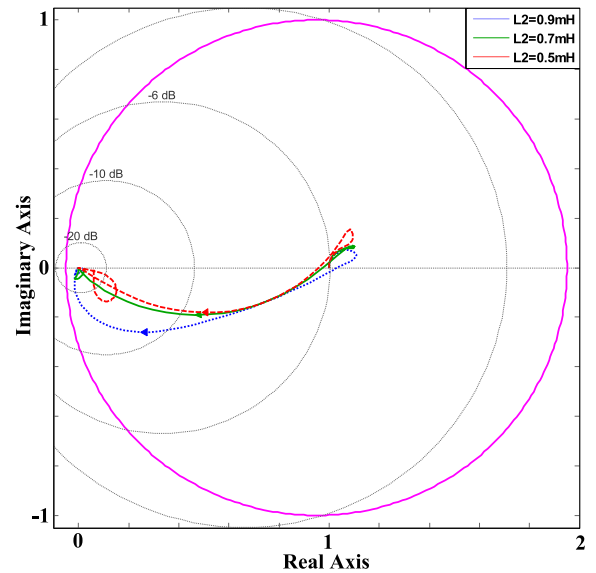
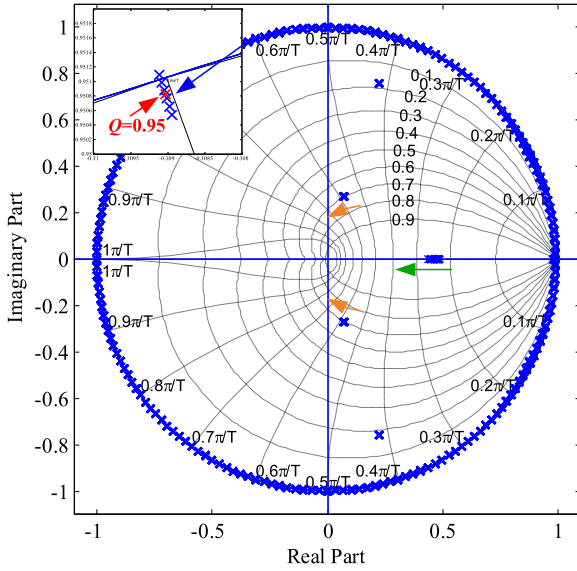


Fig. 15. Trajectory of $S(e^{j\omega T})G_{cl}(e^{j\omega T})$ for different inductors.

changes from 1 to 100, the system poles move to the left boundary of the unit circle. K_i has little effect on system stability, but when K_i is too large the pole will move outside the unit circle, thereby making the system unstable. As shown in Fig. 17, when $K_p = .025$ and $K_i = 10$ are selected, the system is stable.

Fig. 16. Closed-loop pole diagram for different Q ($e^{j\omega T}$).

4.2. Analysis of steady-state accuracy

From (18), the relationship $G_r(z)$ between output error and reference can be expressed as follows

$$G_r(z) = \frac{(1 - G_{cl}(z))(1 - Q(z)z^{-N})}{1 - z^{-N}(Q(z) - S(z)G_{cl}(z))} \quad (20)$$

The error frequency characteristic can be seen in Fig. 18. The amplitude attenuation gains of the three controllers are -28 dB, -30 dB and -60 dB at the fundamental frequency, respectively. With the increase of the frequency, the error amplitude is sharply attenuated for RC with feedforward. It can be seen that the RC has excellent steady-state performance. Compared to the methods with no reference feedforward, the RC method with

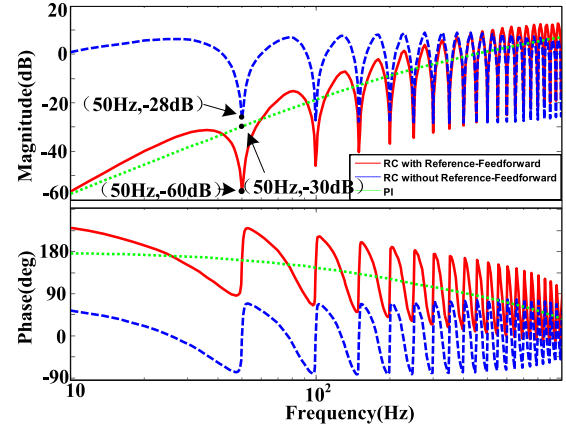


Fig. 18. Error frequency characteristic.

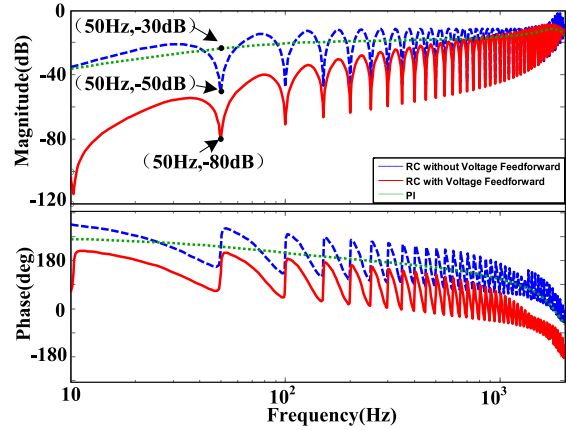
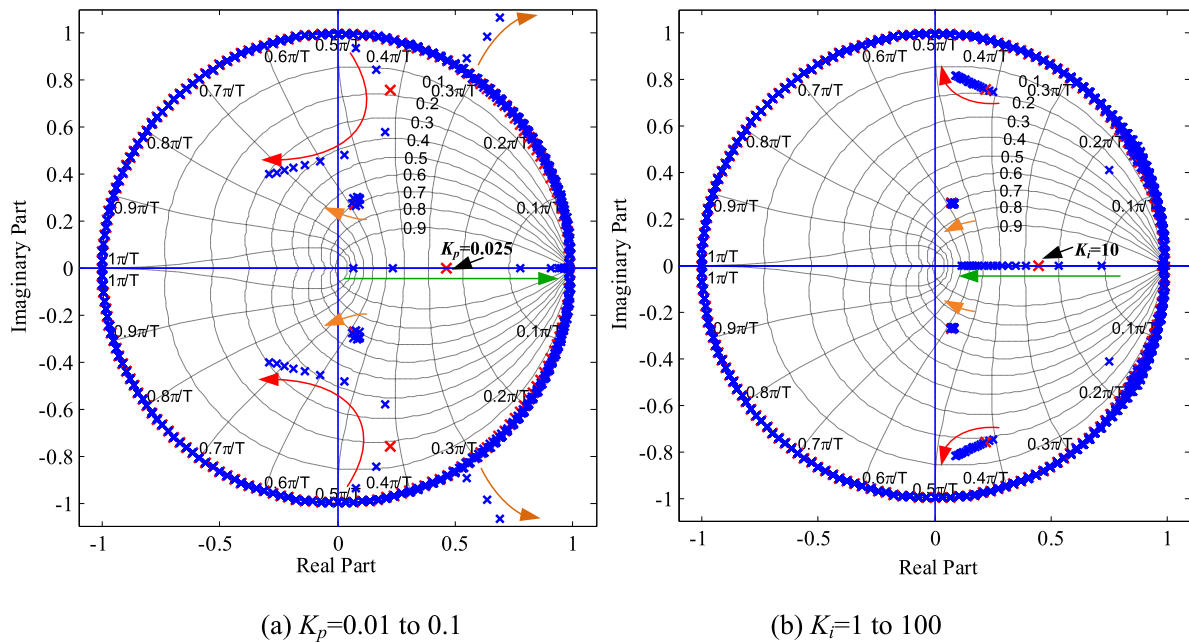
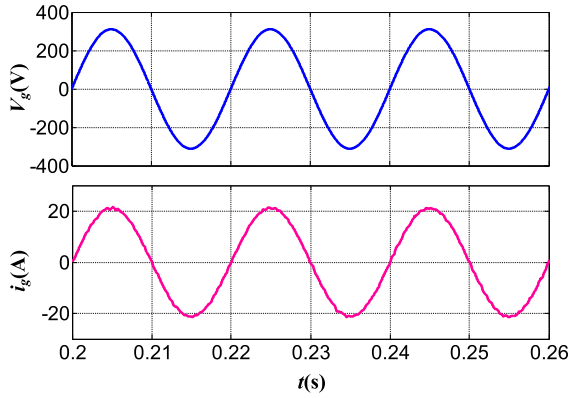
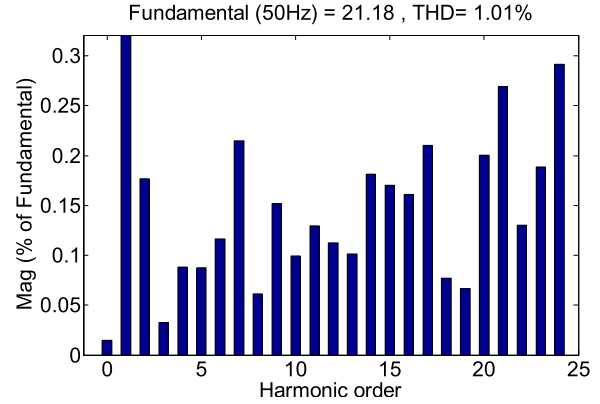


Fig. 19. Frequency characteristic of grid voltage anti-disturbance.

Fig. 17. Closed-loop pole diagram for different K_p or K_i .



(a) steady-state waveforms



(b) the fast Fourier transform (FFT) of the grid current

Fig. 20. The simulation results in steady state.

feedforward has a better attenuation performance because there is an additional polynomial $1-G_{cl}(z)$ in the numerator of the transfer function, and it is close to zero after the inner loop correction. At the same time, it also shows that the inner loop unit can reduce the burden of RC and improve the steady state accuracy of the system.

4.3. Analysis of grid voltage harmonic suppression capability

In grid-connected inverter systems, the main source of

disturbance comes from grid-voltage harmonics. It is very important, therefore, to suppress the harmonic components of the grid voltage to improve the performance of the grid-connected inverter. From (18), the relationship $G_v(z)$ between output error and grid voltage can be expressed as follows

$$G_v(z) = \frac{(1 - Q(z)z^{-N})(G_2(z) - G_{fd}(z)G_p(z))}{1 - z^{-N}(Q(z) - S(z)G_{cl}(z))} \quad (21)$$

Fig. 19 shows the frequency characteristics of the grid voltage anti-disturbance in the PI control method, the RC method without

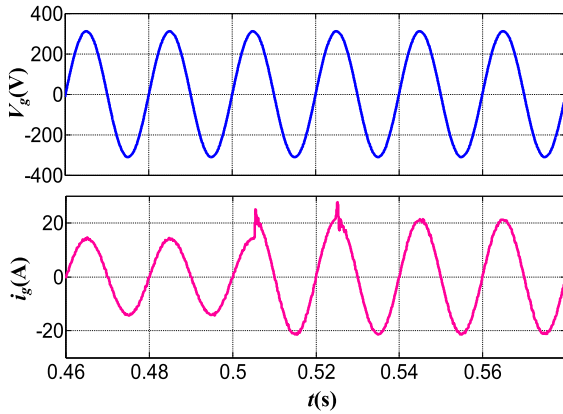


Fig. 21. The simulation results in a dynamic state.

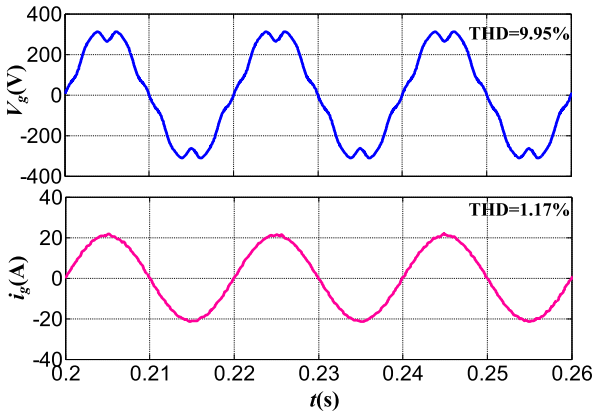


Fig. 22. The simulation results in a grid voltage with harmonics.

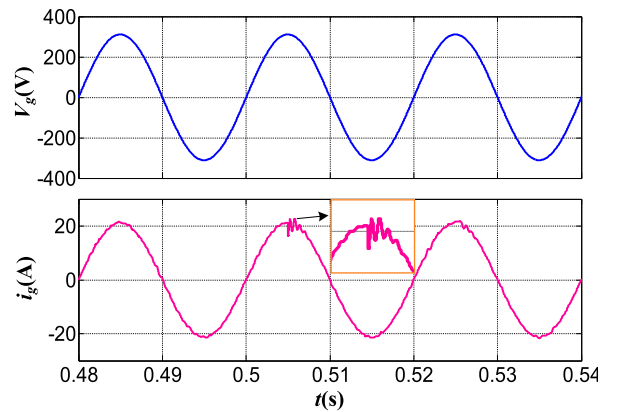


Fig. 23. The simulation results when the grid impedance is changed.

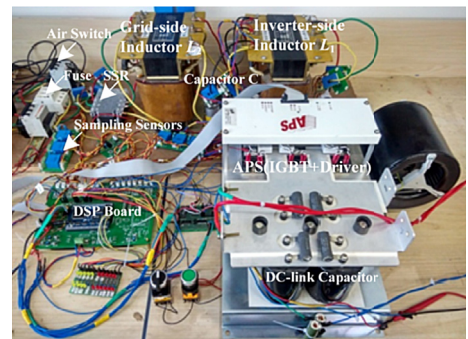
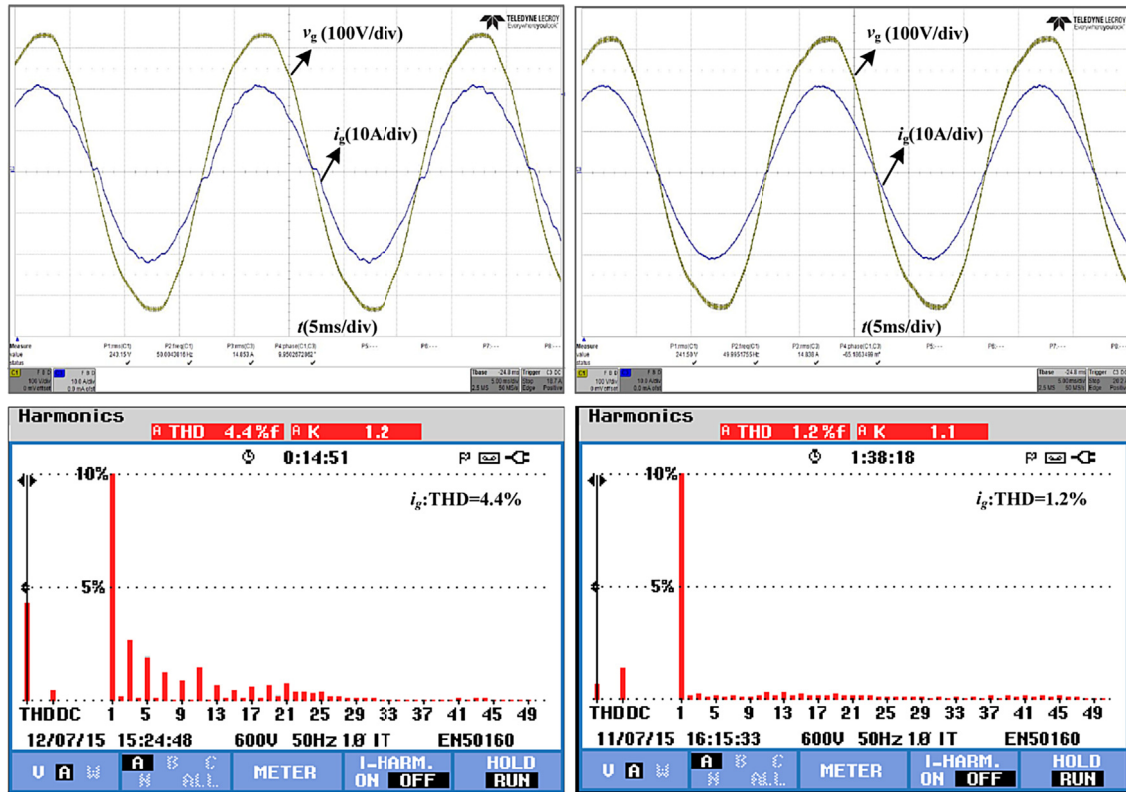
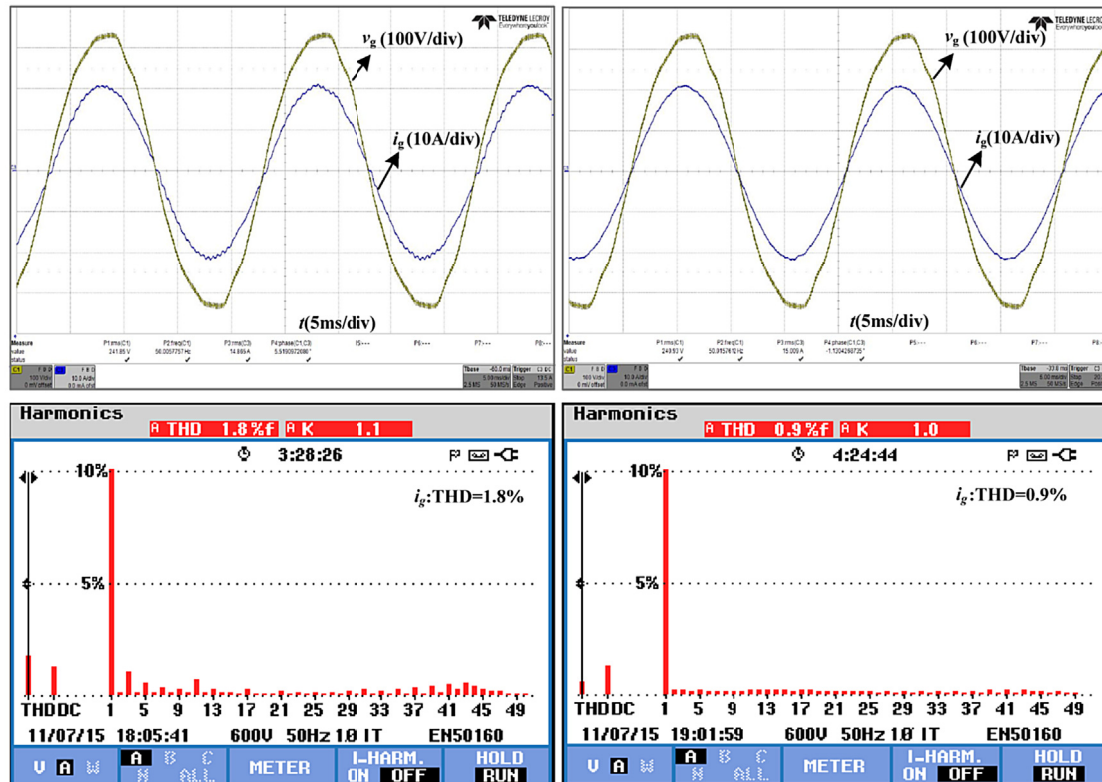


Fig. 24. The prototype of the single-phase grid-connected inverter with an LCL filter.



a) With PI control.

(b) With single-loop RC.



(c) With the repetitive dual-loop control based on inverter current feedback. (d) With the proposed repetitive dual-loop control.

Fig. 25. Experimental results of the system in a steady state.

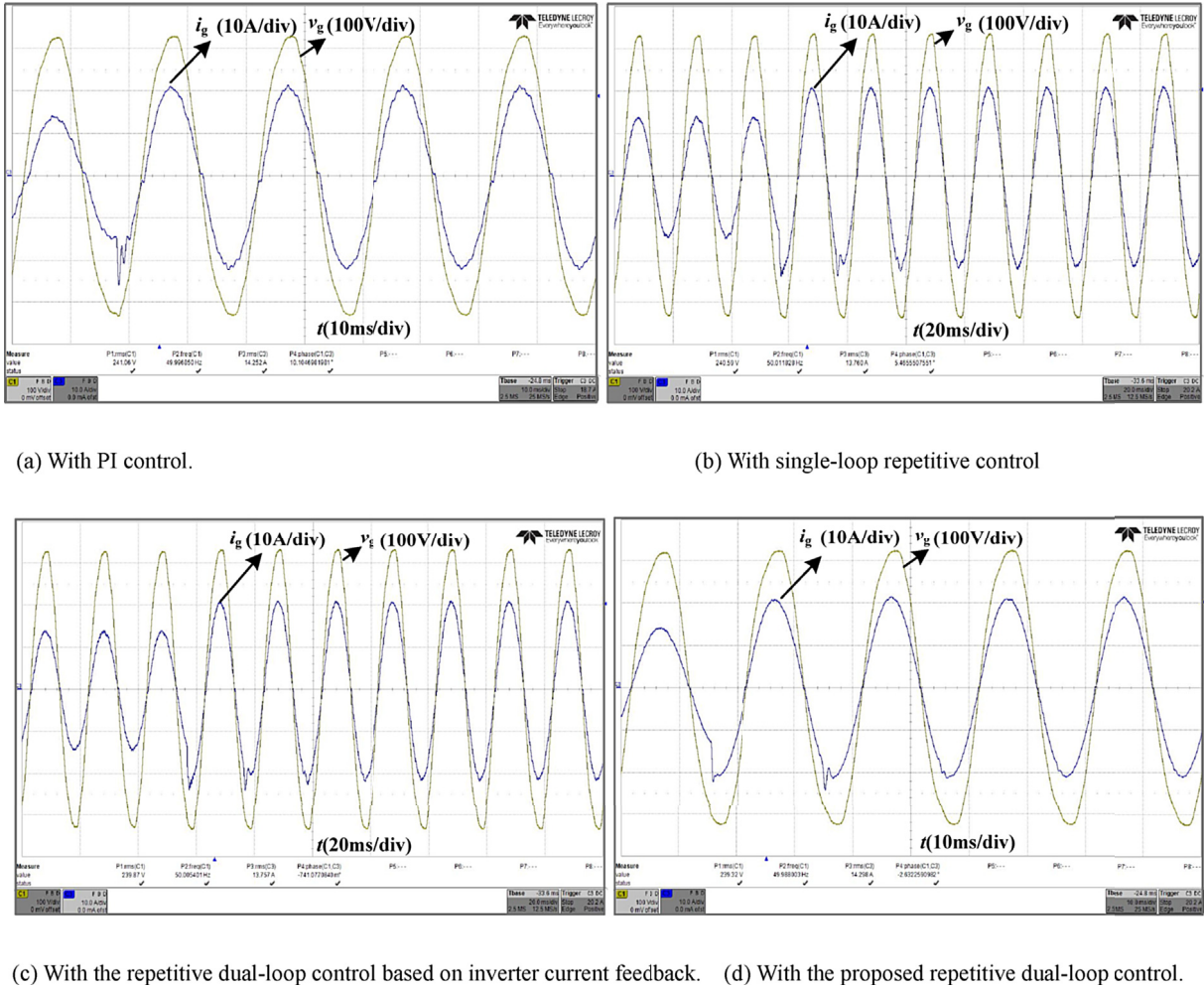


Fig. 26. Experimental results of the system in a dynamic state.

grid-voltage feedforward, and the RC method with grid-voltage feedforward. Fig. 19 shows the attenuation gains at the fundamental frequency are -30 dB and -50 dB, respectively, under the two different conditions without the grid-voltage feedback. After adding the grid-voltage feedforward, the attenuation gain is very small at the fundamental and harmonic frequencies, in which the minimum gain can be up to -80 dB. Obviously, the system error caused by grid-voltage disturbance is further weakened. It follows that the proposed method has better harmonic suppression ability, mainly because $G_2(z) \cdot G_{fd}(z) \cdot G_p(z)$ is extremely close to zero based on (16).

4.4. Analysis of the anti-impedance perturbation capability

With the increasing access of DPGS, the grid voltage can no longer be considered as an ideal sinusoidal voltage source. Actually, it contains variable grid impedance and a large number of harmonics, which presents the weak grid feature [30]. When the inverter is operating under a weak grid condition, the stability problem can be aroused by the grid impedance variation. In this paper, the grid impedance variation is simulated by changing the inductor L_2 . Seen in Fig. 15, the system is still stable, whether the impedance becomes larger or smaller. It can be seen from this that the system is robust.

5. Simulation and experimental validation

5.1. Simulation results

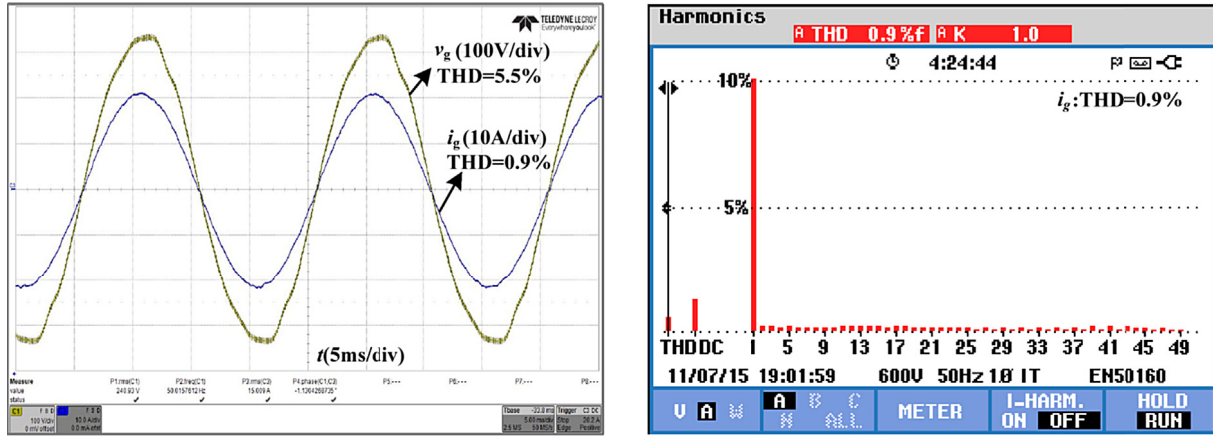
Simulations for the control methods proposed in this paper were carried out using Simulink in MATLAB software. Table 1 contains the parameters used to conduct these simulations. The simulation results are shown in Figs. 20–23.

Fig. 20(a) gives the steady-state waveforms for grid-connected current reference $I_g^* = 15$ A (RMS). In Fig. 20(b) it can be seen that fundamental amplitude of the grid-connected current is 21.18A, the steady-state error is 0.15%, and THD is 1.01%.

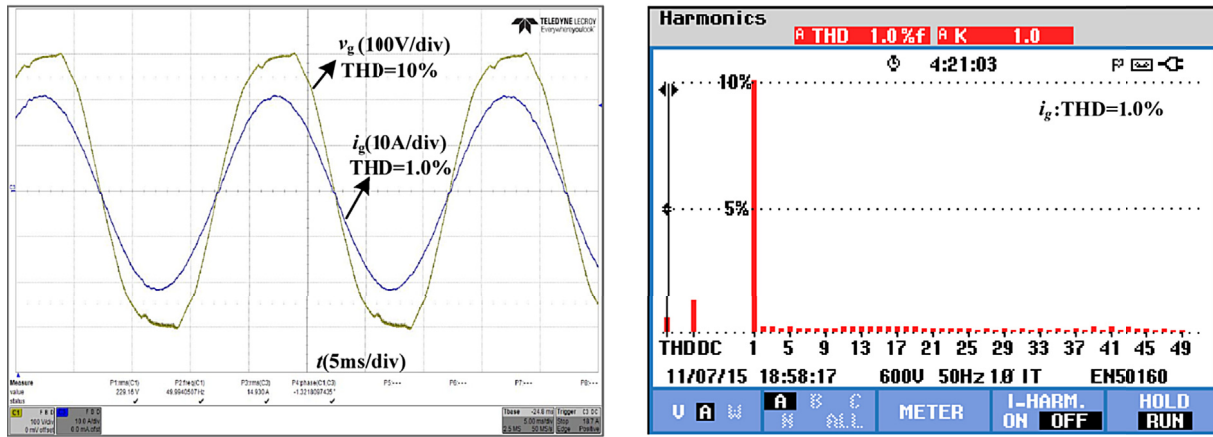
When the grid-connected current reference steps from 10A to 15A at 0.505 s, the dynamic response is as shown in Fig. 21. The response time is within 1 ms, and there is about a 20% overshoot. The system is able to enter the steady state only after two periods.

Fig. 22 gives the simulation result when 9% 5th, 3% 7th and 3% 11th order harmonics are contained in the grid voltage. From Fig. 22, it can be seen that the grid-connected current THD is 1.17% when the grid voltage contains 9.95% harmonics, which shows excellent harmonic suppression ability.

Fig. 23 gives the simulation result when the grid impedance is changed from 0.7 mH to 0.9 mH at 0.505 s. The grid-connected oscillations can only be seen during the step transient. After that, the system is re-stabilized, indicating strong stability.



(a) When the THD of grid voltage is 5.5%.



(b) When the THD of grid voltage is 10%.

Fig. 27. Experimental results of grid-voltage harmonic-suppression capability.

In summary, the simulation results verify the positive characteristics of the proposed control method.

5.2. Experimental results

A 3.3 kW experimental setup is shown in Fig. 24. The prototype includes Applied Power Systems (APS) which consists of an IGBT inverter bridge and PWM driver, a dc-link, an LC filter, a grid interface inductor L_2 , a dc-link capacitor, air switches, a solid state relay (SSR) to disconnect the inverter from the grid when failures happen, voltage and current sensors, a fuse, and a DSP control board with sampling circuits.

The experiment was carried out based on a repetitive dual-loop control scheme. Moreover, the experiments of the PI control, single loop RC in Ref. [18], and the repetitive dual-loop control based on inverter current feedback in Ref. [23] were also presented for comparison to validate the effectiveness of the proposed control method in practical applications.

5.2.1. Steady performance

The contrasting experiments of steady performance of four control methods were carried out under the conditions where the current reference i_g^* was set at 15A and the THD of grid voltage was 5.5%, as shown in Fig. 25.

Shown in Fig. 25(a), using the PI control method, the RMS of the

grid-connected current is 14.853A, but the low-order harmonic content is very high, and the THD is up to 4.4%. Due to the influence of the dead zone, the zero crossing distortion is more serious. As a result, the PI control method is of limited gain and it is impossible to have zero steady-state error tracking for sinusoidal reference. Fig. 25(b) shows that by using the single-loop RC method in Ref. [18], the RMS of the grid-connected current is 14.838A, the grid-connected current harmonic content was reduced, and the THD is only 1.2%. This shows that cross-distortion of the single-loop current feedback control has been greatly improved, and the steady-state performance of the system becomes better than that achieved using the PI controller, which indicates that the RC method can track the sinusoidal reference well. Fig. 25(c) shows that by using the repetitive dual-loop control method in Ref. [23], the RMS of the grid-connected current is 14.865A, the high-order harmonic content is near the 2.21 kHz resonant frequency is also very high, and the THD is up to 1.8%. When the inverter-side current with the high frequency content is used as feedback, the control performance becomes poor. Moreover, this method belongs to indirect control, in which the steady-state accuracy is not high. Shown in Fig. 25(d), harmonic content is very low and the output is 15.009A when using the dual closed-loop control method proposed in this paper. The steady-state error is 0.18%. It is clear that the proposed control method has better steady performance than the other methods, and the THD is only 0.9%. The complete control

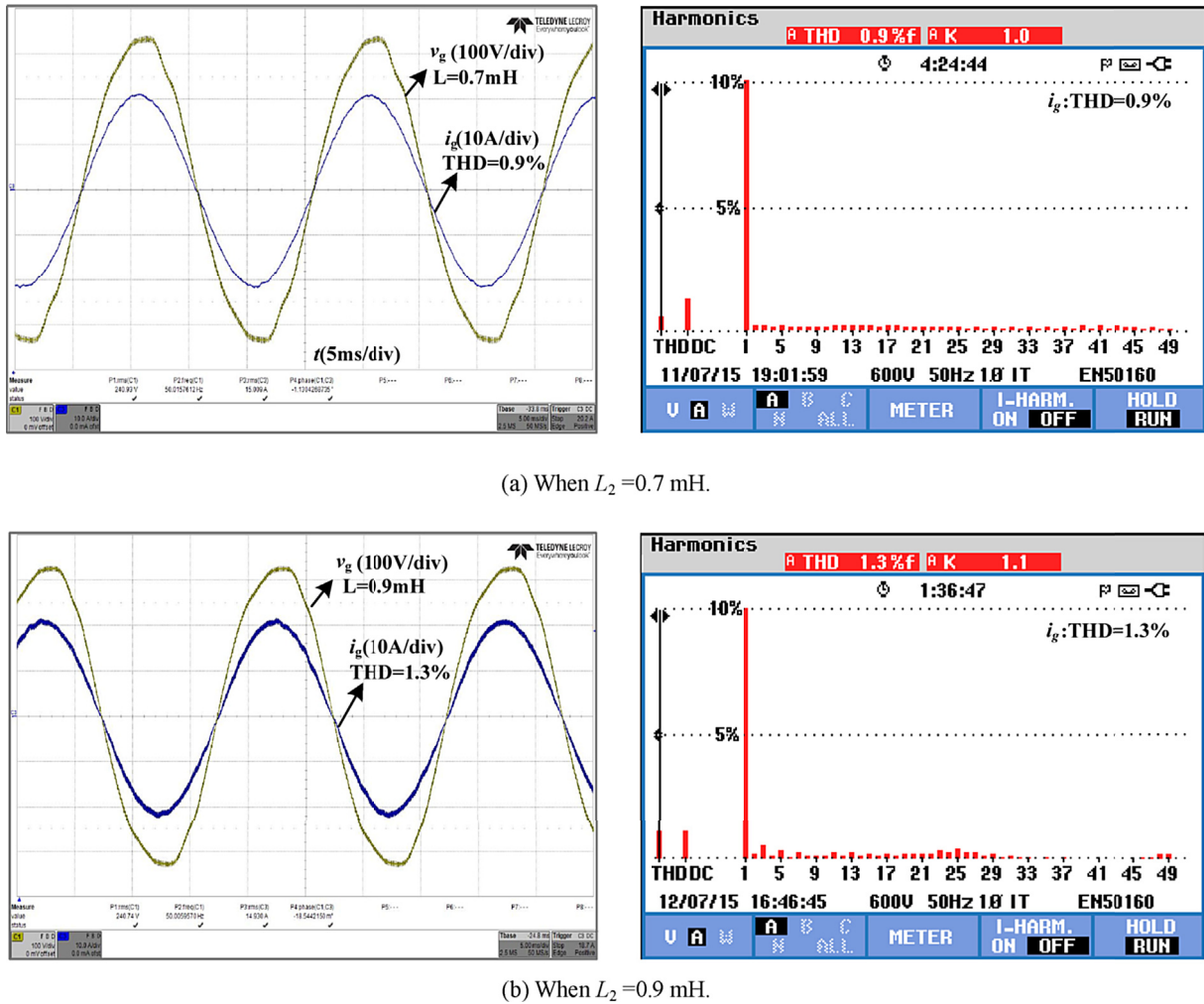


Fig. 28. Experimental results of anti-impedance perturbation capability.

system was calibrated to the characteristics of zero phase-shift at the medium and low frequency ranges and high gain attenuation in the high frequency range. In addition, using the dual-loop control method, the direct control of the grid-connected current can be realized, and good tracking performance for the given and steady-state accuracy can be achieved, which verifies the correctness of the theoretical analysis.

5.2.2. Dynamic performance

Dynamic response capability is one of the most important indexes of a control system. When the reference current is suddenly changed from 10A to 15A, the system response waveforms are shown in Fig. 26.

From Fig. 26(a), it can be seen that in the single-loop PI control method, the system is able to enter the steady state after a fundamental period, and the dynamic response is relatively fast. However, there is a large distortion in the grid-connected current. Fig. 26(b) shows that by using the single-loop RC method in Ref. [18], although the current waveform is better than that in Fig. 26(a), the system still required four fundamental periods to eliminate RC error accumulation and reach a steady state. Because of an inherent period-delay from the RC method, the system cannot respond in the first fundamental period, and the corresponding compensation effect is greatly reduced. From Fig. 26(c), it is obvious that the steady-state accuracy was improved by the repetitive dual-

loop control method in Ref. [23], but four fundamental periods were still needed to eliminate the switching point error and stabilize the system. Fig. 26(d) shows that, compared to the above situation, in the proposed scheme the grid current quickly tracks current reference, and there is a smaller overshoot. The system is able to enter the steady state only after two periods because of the use of the reference feedforward and inner loop units. When the current reference is suddenly changed, the inner loop unit was able to quickly respond to the current reference, which guarantees the compensation efficiency to a certain extent. In the next period, the RC outer loop was only there to regulate the error on the basis of the inner loop, which will greatly reduce the adjustment period. Therefore, the experimental results show that the proposed scheme not only guarantees the steady-state accuracy of compensation, but also has a fast dynamic response.

5.2.3. Grid voltage harmonic-suppression capability

Harmonic-suppression capability is another important index of the control system. To simulate the variation of power quality, large capacity rectifier loads were connected to the power grid. Fig. 27 shows the system responses when the THD of grid voltage is 5.5% (without rectifier loads connection) and 10% (with rectifier loads connection). It can be seen that with the deterioration of the power grid, the THD of the grid-connected current only changes from 0.9% to 1.0%, which shows excellent harmonic suppression ability. This

can be attributed to the use of the grid-voltage feedforward loop and RC method. The high gains of the RC method can enhance the anti-harmonic ability of grid voltage, and the grid-voltage feedforward control loop can effectively reduce the current harmonics caused by grid voltage distortion.

5.2.4. Anti-impedance perturbation capability of grid voltage

The grid interface inductor L_2 was changed from 0.7 mH to 0.9 mH to simulate the variation of grid impedance. These experimental waveforms are shown in Fig. 28. Fig. 28 shows that when the grid-side inductor is 0.9 mH, due to the shift of the resonance peak point, the harmonic content is mainly concentrated in the 23th, 25th and 27th harmonic. However, the system is still stable, and THD merely changes from 0.9% to 1.3%. When the system works in a weak grid, the high gain of the double-loop control method suppresses the grid impedance disturbance. As a result, it has been shown that the system has strong robustness to grid impedance.

6. Conclusion

This paper proposed a grid-side current feedback based repetitive dual-loop control method for grid-connected inverters with an LCL filter. The corresponding digital PI inner-loop feedback and the RC outer-loop feedback were designed. Through the study given in this paper, the following conclusions can be deduced:

- (1) The active damping method based on digital PI was proposed to realize the stability of the LCL filter. Using the inherent delay characteristics of digital control, the system damping and amplitude margin can be increased and the stability of the system can be improved. Moreover, PI control also ensures the fast transient characteristics. RC can realize a zero steady-state error by high gain attenuation at the multiples of the fundamental frequency. Meanwhile, due to the PI inner loop, the repetitive outer loop design process is simpler. Compared to traditional damping methods, the proposed RC dual-loop control method has the advantages of a simple structure, high steady-state accuracy, fast response, and low cost.
- (2) Based on the RC dual-loop control method, a reference current feedforward link and grid-voltage feedforward link were designed to suppress the disturbance of power grid harmonics and impedance, which enhances the system dynamic response.
- (3) The judging formula for stability was derived and the anti-disturbance performance was analyzed. The analysis result showed that when the power grid impedance is fluctuating at about ± 0.2 mH, the system can still be stable; moreover, the system has very high open-loop gain and voltage attenuation gain, which reaches 26 dB and -80 dB in the fundamental frequency, respectively.
- (4) The simulation results verify the positive characteristics of the proposed control method. The experiments herein were carried out on a 3.3 kVA single-phase grid-connected inverter experimental setup. The experimental results showed that the grid current THD is only 0.9%, far lower than 5% of IEEE Standard 1547; the system can directly track the reference current in two periods. When the grid voltage distortion occurs, output current THD change is smaller (0.9%–1.0%); when the grid impedance appears fluctuations (± 0.2 mH), output current THD is still low (only changing from 0.9% to 1.3%). It was therefore proved that systems using this dual-loop control method will have good steady-state performance, fast dynamic response, and strong robustness.

Acknowledgments

This paper and its related research work were supported by National Natural Science Foundation of China (project No. 51577123), the Program for Returned Personnel Studying Abroad in Shanxi Province (2015) and Open fund of Key Laboratory of Mining Electrical Equipment and Intelligent Control (project No. MEI201602).

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